

```
In [82]: import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_excel('Department Awareness Survey (Responses).xlsx')
```

```
In [83]: df.head()
```

```
Out[83]:
```

	Timestamp	Registration Number	Email	Job role that you are interested in	What is your current role?
0	2026-06-22 08:49:31.362	2547123	jiyaelza.jabi@mca.christuniversity.in	Software Engineer	
1	2026-06-22 08:49:50.520	2547122	jinishaleema.rosario@mca.christuniversity.in	Data Engineer	7
2	2026-06-22 08:50:44.521	2547101	rajeev.chandar@mca.christuniversity.in	Software Engineer	
3	2026-06-22 08:51:00.666	2547156	sounak.chakraborty@mca.christuniversity.in	Data Scientist	
4	2026-06-22 08:51:42.357	2547148	samar.subhash@mca.christuniversity.in	Software Engineer	

```
In [84]: print("Rows, Columns:", df.shape)
```

```
Rows, Columns: (54, 15)
```

```
In [85]: print("The datatypes of the columns are:")
print(df.dtypes)
```

The datatypes of the columns are:

Timestamp

datetime64[ns]

Registration Number

object

Email

object

Job role that you are interested in\n

object

What is the minimum salary of students placed through campus (In LPA..respond as a number) object

What is the maximum salary of students placed through campus (In LPA..respond as a number) object

What is the median salary of students placed through campus (In LPA..respond as a number) object

Which is the highest paying company

object

Rate your contribution towards extra curricular activities

int64

Rate your technical competencies

int64

What are your package expectations (LPA)

object

your CIA % of last semester

object

your GPA of last semester

object

Your maximum attendance % till last semester

object

Internships Interests

object

dtype: object

```
In [86]: cols = [
    'your CIA % of last semester ',
    'your GPA of last semester',
    'Your maximum attendance % till last semester'
]

for col in cols:
    df[col] = pd.to_numeric(
        df[col],
        errors='coerce'
    )
```

```
In [87]: df[cols].dtypes
```

```
Out[87]: your CIA % of last semester    float64
your GPA of last semester              float64
Your maximum attendance % till last semester  float64
dtype: object
```

```
In [88]: df.isnull().sum()
```

```
Out[88]: Timestamp
0
Registration Number
0
Email
0
Job role that you are interested in\n
0
What is the minimum salary of students placed through campus (In LPA..respond a
s a number)    0
What is the maximum salary of students placed through campus (In LPA..respond a
s a number)    0
What is the median salary of students placed through campus (In LPA..respond a
s a number)    0
Which is the highest paying company
0
Rate your contribution towards extra curricular activities
0
Rate your technical competencies
0
What are your package expectations (LPA)
0
your CIA % of last semester
1
your GPA of last semester
1
Your maximum attendance % till last semester
6
Internships Interests
0
dtype: int64
```

```
In [89]: df[[
    'your CIA % of last semester ',
    'your GPA of last semester',
    'Your maximum attendance % till last semester'
]].head(20)
```

Out[89]:

	your CIA % of last semester	your GPA of last semester	Your maximum attendance % till last semester
0	78.0000	3.24	NaN
1	64.7800	3.20	NaN
2	80.0000	3.60	NaN
3	70.0000	3.20	92.0000
4	68.0000	3.40	85.0000
5	0.7000	3.50	0.8700
6	0.8800	3.70	NaN
7	3.4500	10.00	90.0000
8	70.0000	3.40	89.0000
9	83.0000	3.89	98.0000
10	70.0000	3.30	97.0000
11	0.7000	3.50	0.9900
12	70.0000	3.48	97.0000
13	64.6700	3.40	95.0000
14	70.0000	2.90	90.0000
15	89.0000	3.67	89.0000
16	0.7111	3.37	0.9425
17	66.0000	2.90	91.0000
18	80.0000	3.60	93.0000
19	78.0000	3.75	95.0000

```
In [90]: df.loc[
    df['your CIA % of last semester '] < 1,
    'your CIA % of last semester '
] *= 100

df.loc[
    df['Your maximum attendance % till last semester'] < 1,
    'Your maximum attendance % till last semester'
] *= 100
```

```
In [91]: df[[
    'your CIA % of last semester ',
    'your GPA of last semester',
    'Your maximum attendance % till last semester'
]].head(20)
```

Out[91]:

	your CIA % of last semester	your GPA of last semester	Your maximum attendance % till last semester
0	78.00	3.24	NaN
1	64.78	3.20	NaN
2	80.00	3.60	NaN
3	70.00	3.20	92.00
4	68.00	3.40	85.00
5	70.00	3.50	87.00
6	88.00	3.70	NaN
7	3.45	10.00	90.00
8	70.00	3.40	89.00
9	83.00	3.89	98.00
10	70.00	3.30	97.00
11	70.00	3.50	99.00
12	70.00	3.48	97.00
13	64.67	3.40	95.00
14	70.00	2.90	90.00
15	89.00	3.67	89.00
16	71.11	3.37	94.25
17	66.00	2.90	91.00
18	80.00	3.60	93.00
19	78.00	3.75	95.00

```
In [92]: df = df[
    (df['your GPA of last semester'] >= 0)
    &
    (df['your GPA of last semester'] <= 4)
]
```

```
In [93]: df = df.dropna(subset=[
    'your CIA % of last semester ',
    'your GPA of last semester',
    'Your maximum attendance % till last semester'
])

df.shape
```

Out[93]: (43, 15)

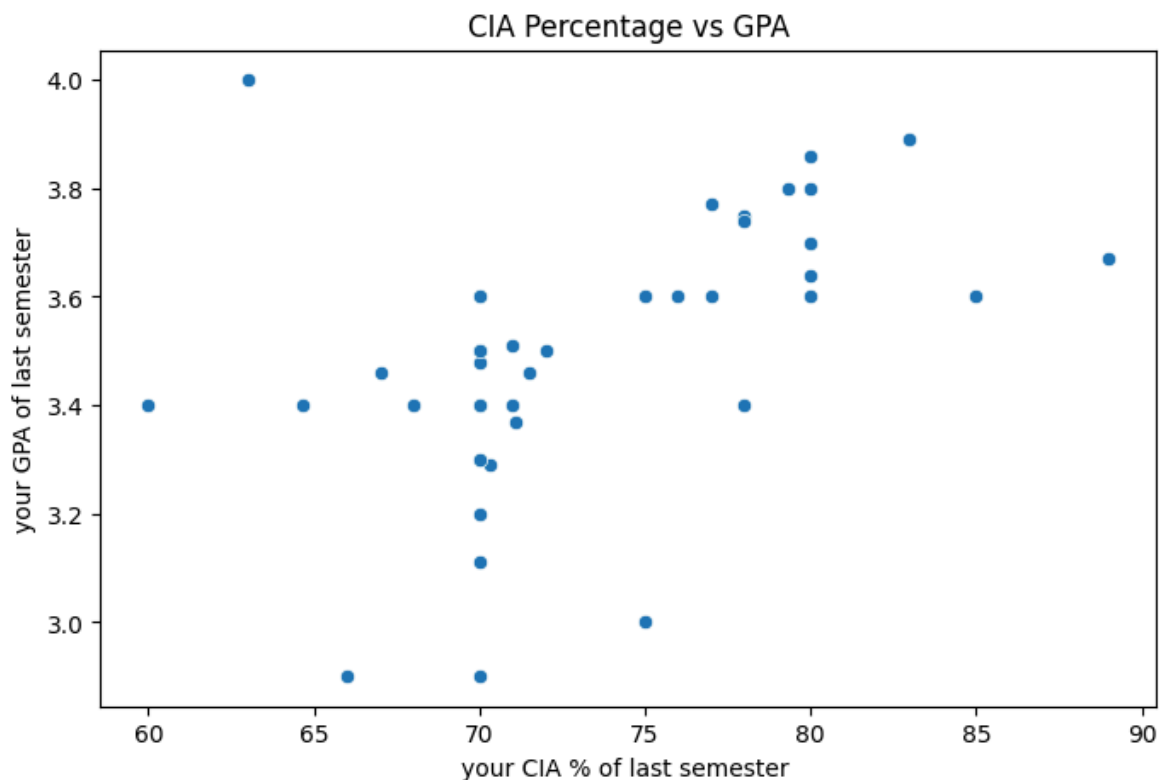
```
In [94]: df = df[df['your CIA % of last semester ' ] >= 20]
```

```
In [95]: df.drop_duplicates()

plt.figure(figsize=(8,5))

sns.scatterplot(
    x='your CIA % of last semester ',
    y='your GPA of last semester',
    data=df
)

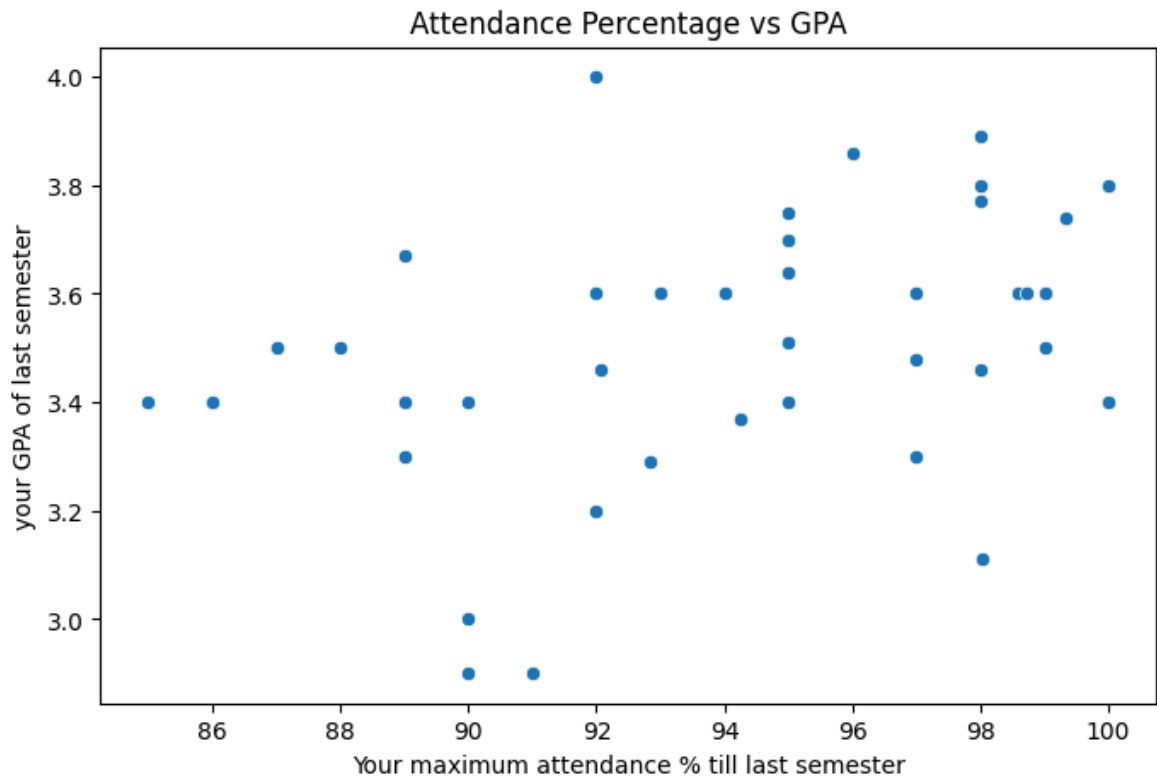
plt.title("CIA Percentage vs GPA")
plt.show()
```



```
In [96]: plt.figure(figsize=(8,5))

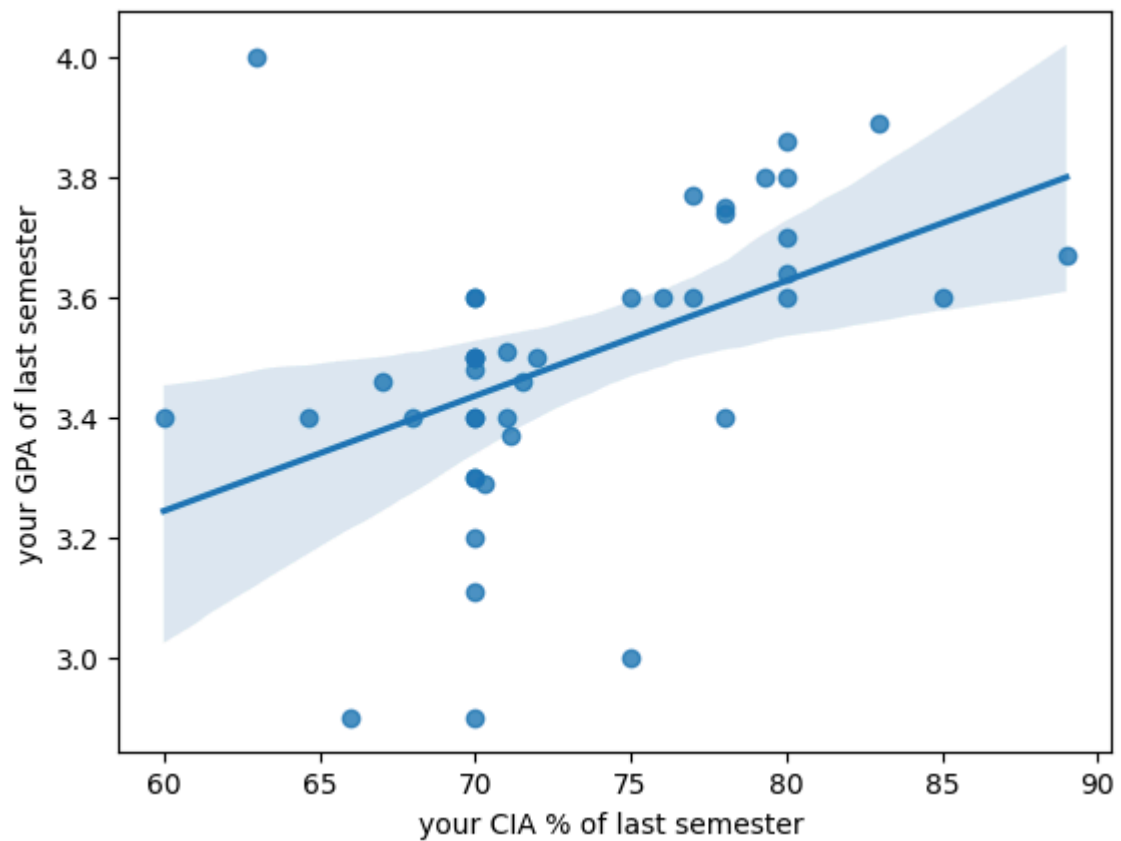
sns.scatterplot(
    x='Your maximum attendance % till last semester',
    y='your GPA of last semester',
    data=df
)

plt.title("Attendance Percentage vs GPA")
plt.show()
```



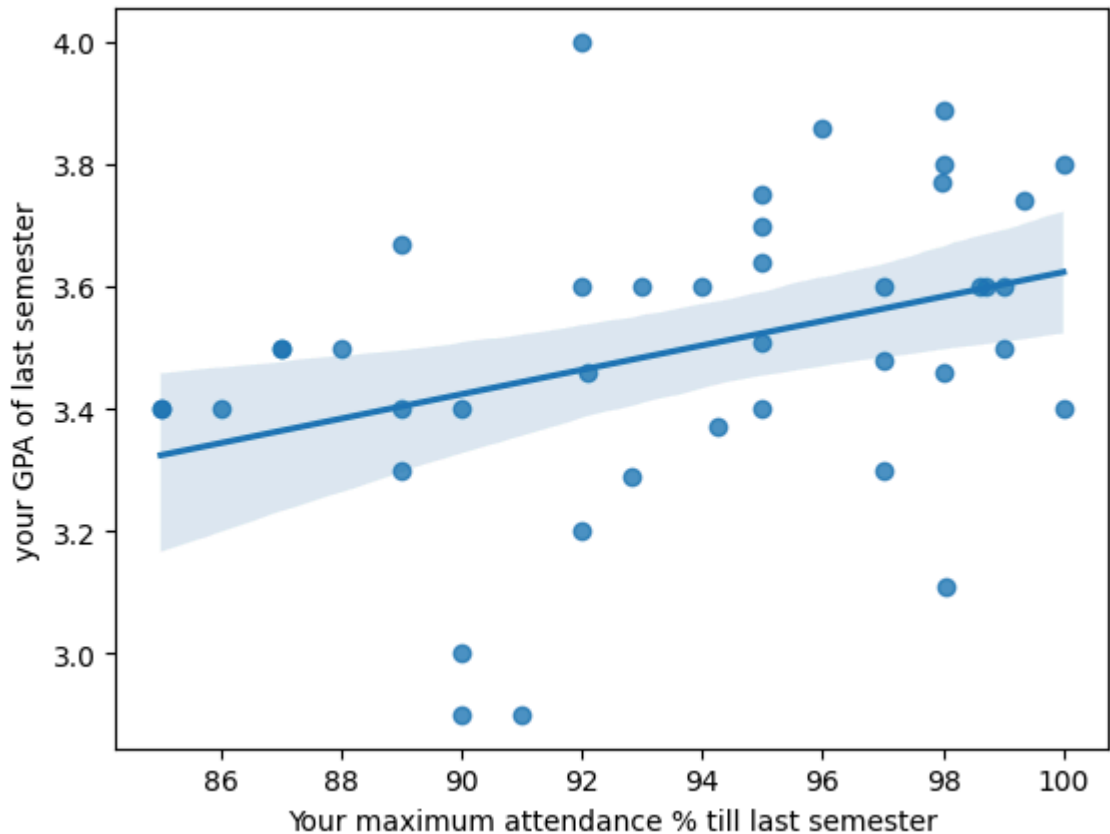
```
In [97]: sns.regplot(
    x='your CIA % of last semester ',
    y='your GPA of last semester',
    data=df
)

plt.show()
```



```
In [98]: sns.regplot(
    x='Your maximum attendance % till last semester',
    y='your GPA of last semester',
    data=df
)

plt.show()
```



```
In [99]: X1 = df[['your CIA % of last semester ']]
    y1 = df['your GPA of last semester']
```

```
In [100... from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X1,
    y1,
    test_size=0.2,
    random_state=42
)
```

```
In [101... from sklearn.linear_model import LinearRegression

model1 = LinearRegression()

model1.fit(X_train, y_train)

print("Slope:", model1.coef_[0])
print("Intercept:", model1.intercept_)

y_pred = model1.predict(X_test)

print(y_pred[:10])
```


Slope: 0.019091285054412023
 Intercept: 2.088885510753385
 [3.52073189 3.61618832 3.32351892 3.44436675 3.67346217 3.53982317
 3.57800574 3.42527546 3.42527546]

Experiment 2: Attendance and GPA

In [102...

```
from sklearn.metrics import r2_score

X2 = df[['Your maximum attendance % till last semester']]
y2 = df['your GPA of last semester']

# Split dataset
X_train2, X_test2, y_train2, y_test2 = train_test_split(
    X2,
    y2,
    test_size=0.2,
    random_state=42
)

# Create model
model2 = LinearRegression()

# Train model
model2.fit(X_train2, y_train2)

# Slope
print("Slope:", model2.coef_[0])

# Intercept
print("Intercept:", model2.intercept_)

# Predictions
y_pred2 = model2.predict(X_test2)

print("Predictions:")
print(y_pred2[:10])

# R2 Score
print("R2 Score:", r2_score(y_test2, y_pred2))
```

Slope: 0.021879948852717795
 Intercept: 1.4352718878120556
 Predictions:
 [3.49198708 3.47010713 3.51386703 3.40446728 3.57950688 3.59504164
 3.62326677 3.33882744 3.44822718]
 R2 Score: -0.21181103746478058